Keyun Cheng

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About

Keyun Cheng is a Ph.D. candidate at the Department of Computer Science and Engineering, The Chinese University of Hong Kong. He is a member of the Applied Distributed System Lab (ADSLab), supervised by Prof. Patrick P. C. Lee. He received the B.Eng. degree in Software Engineering from Sun Yat-Sen University in 2018, the M.Sc. degree in Computer Science from The Chinese University of Hong Kong in 2019. His research interests include distributed storage and erasure coding.

Education

The Chinese University of Hong Kong, Hong Kong — Dept. of Computer Science and Engineering
Ph.D. in Computer Science and Engineering — *Aug. 2020 - present*

- GPA: N/A; Ranking: N/A

The Chinese University of Hong Kong, Hong Kong — Dept. of Computer Science and Engineering
M.Sc. in Computer Science — *Aug. 2018 - Nov. 2019*

- GPA: 3.80/4.00; Ranking: Top 3

Sun Yat-Sen University, Guangzhou, China — School of Data and Computer Science
B.Eng. in Software Engineering — *Aug. 2014 - June. 2018*

- GPA: 3.8/4.0; Ranking: 31/122

Publications

1. **Keyun Cheng**, Huancheng Puyang, Xiaolu Li, Patrick P. C. Lee, Yuchong Hu, Jie Li, and Ting-Yi Wu.
   Toward Load-Balanced Redundancy Transitioning for Erasure-Coded Storage.
   In submission to Transactions on Parallel and Distributed Systems (TPDS) (CCF-A).
2. Zhirong Shen, Yuhui Cai, **Keyun Cheng**, Patrick P. C. Lee, Xiaolu Li, Yuchong Hu, and Jiwu Shu.
   A Survey of the Past, Present, and Future of Erasure Coding for Storage Systems.
   ACM Transactions on Storage (TOS), 21(1), pp. 4:1-4:39, January 2025. (CCF-A).
3. **Keyun Cheng**, Si Wu, Xiaolu Li, and Patrick P. C. Lee.
   Harmonizing Repair and Maintenance in LRC-Coded Storage.
   Proceedings of the 43rd International Symposium on Reliable Distributed Systems (SRDS 2024), Charlotte, NC, USA, September 2024. (CCF-B)
4. Xiaolu Li, **Keyun Cheng**, Kaicheng Tang, Patrick P. C. Lee, Yuchong Hu, Dan Feng, Jie Li, and Ting-Yi Wu.
   ParaRC: Embracing Sub-Packetization for Repair Parallelization in MSR-Coded Storage.
   Proceedings of the 21st USENIX Conference on File and Storage Technologies (FAST 2023), Santa Clara, CA, US, February 2023. (CCF-A)
5. Kaicheng Tang, **Keyun Cheng**, Helen H. W. Chan, Xiaolu Li, Patrick P. C. Lee, Yuchong Hu, Jie Li, and Ting-Yi Wu.
   Balancing Repair Bandwidth and Sub-packetization in Erasure-Coded Storage via Elastic Transformation.
   Proceedings of IEEE International Conference on Computer Communications (INFOCOM 2023), New York, US, May 2023. (CCF-A)

6. Xiaolu Li, **Keyun Cheng**, Zhirong Shen, and Patrick P. C. Lee
Fast Predictive Repair in Erasure-Coded Storage: Analysis, Design, and Implementation.
IEEE Transactions on Parallel and Distributed Systems (TPDS), 33(12), pp. 3400-3414, December 2022.
(An earlier version appeared in DSN 2019) (CCF-A)
7. Yueming Jin, **Keyun Cheng**, Qi Dou, Pheng Ann Heng.
Incorporating Temporal Prior from Motion Flow for Instrument Segmentation in Minimally Invasive Surgery Video.
Medical Image Computing and Computer Assisted Intervention (MICCAI), 2019. (CCF-B) **(Oral)**

Academic Activities

Reviewer

Conference/Journal

- USENIX Conference on File and Storage Technologies (FAST) 2025
- IEEE International Symposium on Information Theory (ISIT) 2024
- ACM Transactions on Storage (TOS)

Teaching Assistant

The Chinese University of Hong Kong — *September 2020 - January 2023*

- CSCI5550 Advanced File and Storage Systems (Instructor: Patrick P. C. Lee): Fall 2023
- CSCI1120B Introduction to Computing Using C++ (Instructor: Yat Chiu Law and King Tim Lam): Fall 2022
- AIST3020 Introduction to Computer Systems (Instructor: Patrick P. C. Lee): Spring 2023, Spring 2022
- CSCI4180 Introduction to Cloud Computing and Storage (Instructor: Patrick P. C. Lee): Fall 2021

Sun Yat-Sen University — *September 2017 - June 2018*

- Probablity Theory and Statistics (Instructor: Weiqi Luo): Fall 2017
- Complexity Analysis for Engineering (Instructor: Weiqi Luo): Spring 2018

Academic Projects

Erasure Coding for Dependable Distributed Storage at Scale — June 2019 - present

Group member — *CUHK (Supervisor: Patrick P. C. Lee)*

- Our research applies erasure coding as a low-cost redundancy mechanism to support large-scale dependable storage in distributed environments, with emphasis on balancing the design trade-offs between storage savings and performance efficiency in practical deployment, backed by theoretical and empirical analysis.

Instrument Segmentation for Robotic Surgical Video — September 2018 - April 2019

Group member — *CUHK (Supervisor: Pheng Ann Heng)*

- Our research considers incorporating instrument motion information in deep learning models for accurate instrument segmentation in robotic surgical video. We design an attention module that infers temporal priors by propagating the prediction results of previous frames to the current frames via optical flow. We adopt an encoder-decoder pyramid segmentation framework based on UNet.

Work and Internship Experience

Research and Development Engineer — August 2019 - July 2020

CU Coding Ltd. — *Shatin, N. T., Hong Kong*

- Focus on research and development on storage technologies, especially on nCloud (a multi-cloud storage system) and iNAS (a document management system).
- Assist system maintenance and software bug fixes.
- Assist project management and provide customer support.

Research and Development Intern — December 2017 – April 2018

Guangzhou Intelligence Communications Technology Co. Ltd. — *Guangzhou, China*

- Focus on research and development on automated image annotation tools based on deep learning.

- Design an automated image annotation tool based on labelImg (an open-source image annotation project) and Mask-RCNN (a novel deep segmentation model). The image annotation tool performs object segmentation and further generates contours and labels for individual objects. The label metadata are stored locally and can be transferred through zeroc-ice (a framework for building networked applications).

Research and Development Intern April 2017 – June 2017
Bigo Technology Pte. Ltd. *Guangzhou, China*

- Focus on research and development on crash log collection and analysis for Android applications.
- Design a crash log upload module in BigoLive (a video streaming app) Android client based on Google Breakpad (a crash-reporting system).
- Design a crash analysis system to collect and analyze crash logs generated by BigoLive Android client.
- Detect and analyze network abnormality from crash logs generated by BigoLive Android client.

Honors and Awards

Postgraduate Studentship, CUHK	2020 - 2024
Dean's List Scholarship, Faculty of Engineering, CUHK	2019
First Prize Winner (top 3) of Distinguished Academic Performance Scholarship, M.Sc. in CS, CUHK	2019
Second Class Scholarship, SYSU	2017
Meritorious Winner of Mathematical Contest in Modeling (MCM)	2017
First Class Scholarship, SYSU	2016
Third Prize of China Undergraduate Mathematical Contest in Modeling (CUMCM)	2016
Honorable Mention of Mathematical Contest in Modeling (MCM)	2016
Third Class Scholarship, SYSU	2015
Third Prize of ACM programming competition in SYSU	2015

Misc

Programming Languages: C, C++, Python, Java
Github: https://github.com/keyuncheng
English Proficiency: CET-4: 577, CET-6: 532, IELTS: 7.0 (L: 8.5 R: 7.0 W: 6.0 S: 5.5)
Hobbies: long-distance running, basketball, guitar